

1) monthly prices $\Rightarrow 100, 108, 103, 115, 110$

$$a) R_t = (P_t - P_{t-1}) / P_{t-1}$$

Month 1 $\rightarrow$ 2

$$R_1 = \frac{108 - 100}{100} = 0.08$$

Month 2 $\rightarrow$ 3

$$R_2 = \frac{103 - 108}{108} = -0.0463$$

Month 3 $\rightarrow$ 4

$$R_3 = \frac{115 - 103}{103} = 0.1165$$

Month 4 $\rightarrow$ 5

$$R_4 = \frac{110 - 115}{115} = -0.0435$$

b) log returns $r_t = \ln(P_t / P_{t-1})$

Month 1

$$\Rightarrow r_1 = \ln\left(\frac{108}{100}\right) = \ln(1.08) \approx 0.0770$$

Month 2

$$\Rightarrow r_2 = \ln\left(\frac{103}{108}\right) \approx -0.0474$$

Month 3

$$r_3 = \ln\left(\frac{115}{103}\right) \approx 0.1102$$

Month 4

$$r_4 = \ln\left(\frac{110}{115}\right) \approx -0.0445$$

c)

Difference:-

$$R_t - r_t$$

$$R_1 - r_1 = 0.0030$$

$$R_2 - r_2 = 0.0011$$

$$R_3 - r_3 = 0.0063$$

$$R_4 - r_4 = 0.0010$$

Least accurate month $\Rightarrow$ Month 3

Because return is biggest (11.65%)

Large returns $\Rightarrow$ more error

Q2 $E[R_1] = 15\%$

$$E[R_2] = 7\%$$

$$\omega_1 = 0.4$$

$$\omega_2 = 0.6$$

a) linear operator $E[ax+by] = a \cdot E[x] + b \cdot E[y]$

$$E[R_p] = E[w_1 R_1 + w_2 R_2] = w_1 E[R_1] + w_2 E[R_2]$$

$$E[R_p] = 0.102 \approx 10.20\%$$

b) No, this result doesn't change.

Q3 $R_p = w_1 R_1 + w_2 R_2$

let $X = w_1 R_1, Y = w_2 R_2$

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y);$$

$$\sigma_p^2 = \text{Var}(w_1 R_1 + w_2 R_2) = \text{Var}(w_1 R_1) + \text{Var}(w_2 R_2) + 2 \text{Cov}(w_1 R_1, w_2 R_2)$$

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

$$\text{Cov}(aX, bY) = ab \text{Cov}(X, Y)$$

$$\sigma_p^2 = w_1^2 \text{Var}(R_1) + w_2^2 \text{Var}(R_2) + 2w_1 w_2 \text{Cov}(R_1, R_2)$$

$$\text{Var}(R_1) = \sigma_1^2$$

$$\text{Var}(R_2) = \sigma_2^2$$

$$\rho_{12} = \frac{\text{Cov}(R_1, R_2)}{\sigma_1 \sigma_2}$$

$$\text{Cov}(R_1, R_2) = \rho_{12} \sigma_1 \sigma_2$$

$$\sigma_p^2 = w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + 2w_1 w_2 \rho_{12} \sigma_1 \sigma_2$$

Q4)

Asset A: $\mu_A = 18\%$ $\sigma_A = 25\%$

Asset B: $\mu_B = 8\%$ $\sigma_B = 12\%$

weights

$$w_A = w_B = 0.5$$

a) Portfolio risk

$$\sigma_p^2 = w_A^2 \sigma_A^2 + w_B^2 \sigma_B^2 + 2w_A w_B \rho \sigma_A \sigma_B$$

for $\rho = +1$:

$$\sigma_p = 0.5(25) + 0.5(12) = 18.5\%$$

for $\rho = 0$:

$$\sigma_p = \sqrt{(0.5)^2(25)^2 + (0.5)^2(12)^2} = 13.87\%$$

for $\rho = -1$:

$$\sigma_p = |0.5(25) - 0.5(12)| = 6.5\%$$

Hence:

$$\rho = +1 \rightarrow 18.5\%$$

$$\rho = 0 \rightarrow 13.87\%$$

$$\rho = -1 \rightarrow 6.5\%$$

b) $\rho = +1$

$$\sigma_p^2 = (\omega_A \sigma_A + \omega_B \sigma_B)^2$$

$$\sigma_p = \omega_A \sigma_A + \omega_B \sigma_B$$

c) $\rho = -1$

$$\omega_A \sigma_A = \omega_B \sigma_B$$

$$\omega_B = 1 - \omega_A$$

$$\omega_A = \frac{\sigma_B}{\sigma_A + \sigma_B} = 0.324$$

$$\omega_A \approx 32.4\%$$

Q5

Combining two assets with $\rho < 1$ reduces portfolio risks because assets do not move perfectly together.

From portfolio variance formula

$$\sigma_p^2 = \omega_1^2 \sigma_1^2 + \omega_2^2 \sigma_2^2 + 2\omega_1 \omega_2 \sigma_1 \sigma_2 \rho$$

the cross term depends on correlation. When $\rho = 1$, risk equals the weighted avg. of individual risks. If $\rho < 1$, the covariance effect is smaller, so movements in one asset partially offset movements in other.

Q6

$$\sigma_1 = 25\% = 0.25$$

$$\sigma_2 = 12\% = 0.12$$

$$\sigma_3 = 4\% = 0.04$$

$$p_{ii} = 1$$

correlations $\Rightarrow$

$$p_{12} = 0.4$$
$$p_{13} = 0.1$$
$$p_{23} = 0.2$$

a) Diagonal: ($i=j$ & $p=1$)

$$\Sigma_{11} = \sigma_1^2 = (0.25)^2 = 0.0625$$

$$\Sigma_{22} = \sigma_2^2 = 0.0144$$

$$\Sigma_{33} = \sigma_3^2 = 0.0016$$

off-diagonals:

$$\Sigma_{12} = \Sigma_{21} = p_{12} \sigma_1 \sigma_2 = 0.012$$

$$\Sigma_{13} = \Sigma_{31} = p_{13} \sigma_1 \sigma_3 = 0.001$$

$$\Sigma_{23} = \Sigma_{32} = p_{23} \sigma_2 \sigma_3 = 0.00096$$

$$\Sigma = \begin{bmatrix} 0.0625 & 0.012 & 0.001 \\ 0.012 & 0.0144 & 0.00096 \\ 0.001 & 0.00096 & 0.0016 \end{bmatrix}$$

since all off-diagonal elts are symmetrically equal $\rightarrow \Sigma$ is symmetrical

Q7:

Risk-free rate $R_f = 4\%$

Sharpe ratio

Portfolio	$E[R_p]$	σ_p
X	14%	22%
Y	9%	10%
Z	20%	35%

a) Sharpe ratio??

$$S_p = \frac{E[R_p] - R_f}{\sigma_p}$$

$$S_x = \frac{E[R_p] - R_f}{\sigma_p} = \frac{14 - 4}{22} = 0.455$$

$$S_y = \frac{9 - 4}{10} = 0.50$$

$$S_z = \frac{20 - 4}{35} = 0.457$$

b) Ranking:

$Y > Z > X$

A rational mean-variance investor prefers portfolio Y

because it has the highest Sharpe ratio.

$\Rightarrow$ Best return per unit risk

c) Although Z has highest absolute return it also has very high risk.

Portfolio Y has lower risk comparatively.

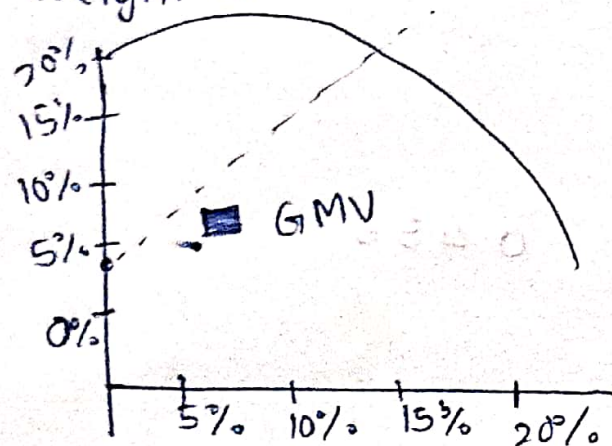
Q8) a)

The Global Minimum Variance (GMV) portfolio:-

The leftmost point.

The portfolio with lowest possible risk across

all weight combinations



It is leftmost point on min-variance frontier because it has minimum risk.

b)

No, this is incorrect. The lower half of min-variance frontier is inefficient because ~~for~~ for same level of risk there exists another portfolio on upper half with higher return. Rational investors always choose portfolios on efficient frontier.

c) The Capital Market line (CML) is constructed by drawing a line from risk free asset to tangency portfolio on efficient frontier.

The slope of CML represents Sharpe Ratio .i.e, excess return earned per unit of risk.

39)

a) 36 observations

$$\sigma_i = 30\%, \quad \hat{\mu}_i = 7\%$$

$$\mu_i = \frac{\sigma_i}{\sqrt{36}} = 5\%$$

b) "Error maximisation" property describes how mean variance optimiser blindly amplifies

small estimation error present in expected returns.

ML & improves prediction quality & reduces instability in portfolio optimisation

40)

$$N = 50$$

$$\frac{N(N+1)}{2} = 1275$$

This is challenging because estimating large no. of cov. terms requires lot of data & ~~est~~ estimation errors.